

ENG-654 · LECTURE 01

Frames, Transformations & D-H Modelling

How do we turn robot geometry into a reproducible kinematic model?

Coordinates become rigid transformations; repeated transformations become a D-H model; numerical checks establish whether that model reproduces the physical robot.

A robot connects a mechanism to decisions about motion

Useful motion requires physical structure, the ability to act and observe, and computation that closes the loop.

Mechanism

Links, joints, and an end effector define what poses the machine can realise.

Actuation

Motors and transmissions turn commanded joint motion into forces and displacement.

Sensing

Encoders and external sensors estimate configuration and the surrounding world.

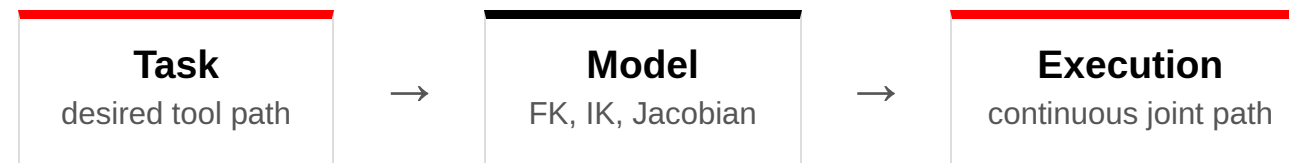
Computation

Models, planning, and feedback choose motions and correct their execution.

This course focuses on the model inside that loop: the geometric structure that connects joint coordinates, workspace poses, singularities, and feasible paths.

Workspace intent must be lifted into joint-space execution

A task is easiest to describe as tool motion in the world, but a robot can execute only admissible joint trajectories.



Workspace describes intent

Tool position and orientation make the task geometrically intuitive.

Joint space exposes feasibility

IK branches, singularities, joint limits, and collisions determine whether and how that intent can be executed.

Planning elements: a task and start state; a kinematic and environment model; validity tests for IK, singularities, joint limits, and collision; and a strategy that selects one continuous feasible lift from start to finish.

Each lecture supplies one layer of kinematics-grounded planning

The sequence moves from reading a robot to reasoning globally about all motions it can perform.

- **01 · Frames and D-H:** build a reproducible forward model.
- **02 · URDF and PoE:** connect files, geometry, and screw motion.
- **03 · 3R inverse kinematics:** expose solution multiplicity.
- **04 · Wrist-partitioned 6R IK:** separate position and orientation.
- **05 · Singularities:** interpret rank loss geometrically.
- **06 · Cuspidality:** study global IK connectivity and joint limits.
- **07 · Workspace path planning:** compare local numerical and global analytical reasoning.
- **08 · Redundant planning:** use the extra degree of freedom to preserve feasible IK branches.

Today begins the chain: a planner cannot reason about a robot until its frames and transformations are unambiguous.

Notation is part of the model

Before calculating, we must distinguish physical objects, coordinate vectors, and the frames in which those coordinates are expressed.

Frames and geometric objects

W is the fixed world—or reference—frame.

B is a moving body frame.

A, B, C denote generic frames when no frame is privileged as world.

P names one physical point in space.

An italic capital names a geometric object or frame; it is not a coordinate value.

Coordinates and transformations

$\mathbf{p}$ is a point-coordinate vector; $\mathbf{v}$ is a direction vector.

In ${}^W\mathbf{p}_P$, the left superscript says “expressed in W ,” while the subscript identifies point P .

A tilde, as in $\tilde{\mathbf{p}}$, marks the four-coordinate homogeneous representation.

${}^W\mathbf{t}_B$ locates the origin of B , expressed in W .

WR_B is the orientation of B expressed in W .

WT_B is the coordinate map from frame B to frame W .

$${}^W\tilde{\mathbf{p}}_P = {}^WT_B {}^B\tilde{\mathbf{p}}_P$$

Coordinates need a reference frame

A list of numbers locates a point only after an origin and a basis have been specified.

The numbers $[0.4, 0.2, 0.1]^T$ do not locate a point by themselves.

We must know the **origin** from which the coordinates are measured.

We must know the **directions of the three basis axes**.

The same physical point therefore has different coordinate vectors in different frames.

Coordinate frame

An origin plus an ordered, right-handed orthonormal basis $(\hat{x}, \hat{y}, \hat{z})$.

$${}^A\mathbf{p}_P \neq {}^B\mathbf{p}_P \quad \text{in general}$$

Reading a rotation matrix

The columns of a rotation matrix are the moving frame's basis axes written in the reference frame.

The matrix ${}^A R_B$ answers: “How are the axes of B oriented when observed from A ?”

$${}^A R_B = \begin{bmatrix} | & | & | \\ {}^A \hat{x}_B & {}^A \hat{y}_B & {}^A \hat{z}_B \\ | & | & | \end{bmatrix}$$

Orthogonal: ${}^A R_B^T {}^A R_B = I$.

Right-handed: $\det({}^A R_B) = +1$.

Inverse: ${}^B R_A = ({}^A R_B)^T$.

Takeaway: interpret the columns geometrically before memorising matrix identities.

Points and directions are different objects

A fourth coordinate lets projective geometry distinguish a located point from a direction with no finite position.

Projective 3-space uses four coordinates, with the last coordinate denoted by w .

$\tilde{\mathbf{p}} = \begin{bmatrix} \mathbf{p} \\ 1 \end{bmatrix}$ is a finite point,

$\tilde{\mathbf{v}} = \begin{bmatrix} \mathbf{v} \\ 0 \end{bmatrix}$ is a direction.

The tilde marks a homogeneous representation; $\mathbf{p}, \mathbf{v} \in \mathbb{R}^3$.

Why “homogeneous”?

Every nonzero scalar multiple represents the same projective point:

$$[x : y : z : w] = [\lambda x : \lambda y : \lambda z : \lambda w], \quad \lambda \neq 0.$$

A finite point has $w \neq 0$ and can be normalised to $w = 1$.

A direction has $w = 0$, so it lies at infinity and has no finite location.

Vector caveat: projective directions are scale-equivalent, but a physical displacement or velocity retains its magnitude. The form $[\mathbf{v}; 0]$ is used so translations do not act on that vector.

Why a rotation matrix cannot translate

Every 3×3 linear map fixes the origin, while a nonzero translation must move it.

A 3×3 rotation matrix R is linear and orthogonal:

$$R^T R = I, \quad R\mathbf{0} = \mathbf{0}.$$

Every linear map must keep the origin fixed.

A translation by $\mathbf{t}$ is affine:

$$\mathbf{p}' = \mathbf{p} + \mathbf{t}, \quad \mathbf{0} \mapsto \mathbf{t}.$$

If $\mathbf{t} \neq \mathbf{0}$, no 3×3 linear—or orthogonal—matrix can perform this operation.

■ We therefore lift 3D coordinates into four dimensions so rotation and translation can act through one linear matrix multiplication.

Deriving the homogeneous transform

Lifting a 3D point to four coordinates converts the affine rotation-plus-translation rule into one linear matrix multiplication.

Let ${}^W\mathbf{t}_B$ be the position of the origin of B , expressed in W . Then

$${}^W\mathbf{p}_P = {}^WR_B {}^B\mathbf{p}_P + {}^W\mathbf{t}_B.$$

Rotation is linear; translation is the added offset.

Append 1 to each point and choose the last row so that the fourth coordinate remains 1:

$$\begin{bmatrix} {}^W\mathbf{p}_P \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} {}^WR_B & {}^W\mathbf{t}_B \\ \mathbf{0}^T & 1 \end{bmatrix}}_{{}^WT_B} \begin{bmatrix} {}^B\mathbf{p}_P \\ 1 \end{bmatrix}.$$

One matrix stores a complete frame pose

The rotation block describes the basis; the translation block locates the origin; the bottom row preserves homogeneous points.

A **homogeneous transformation** is the 4×4 linear representation of a 3D affine rigid transformation acting on homogeneous coordinates.

$${}^A T_B = \begin{bmatrix} {}^A R_B & {}^A \mathbf{t}_B \\ \mathbf{0}^T & 1 \end{bmatrix}.$$

Rotation: how the axes of B are oriented when seen from A.

Translation: where the origin of B is located when measured in A.

Bottom row: lets rotation and translation be composed by one matrix product.

A frame transform changes coordinates, not geometry

Multiplying by ${}^A T_B$ re-expresses the same point from frame B in frame A .

If the coordinates of P are known in frame B , then

$$\begin{bmatrix} {}^A \mathbf{p}_P \\ 1 \end{bmatrix} = {}^A T_B \begin{bmatrix} {}^B \mathbf{p}_P \\ 1 \end{bmatrix}.$$

The coordinate vector changes.

The physical point does not.

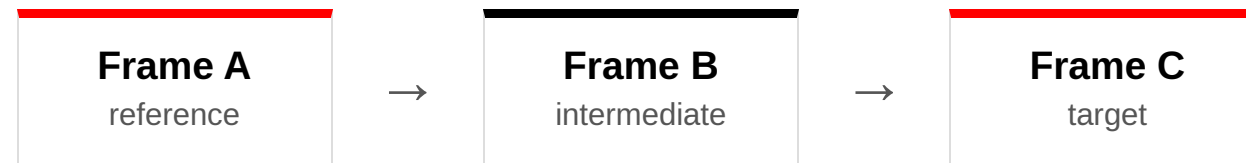
This distinction is essential when reading robot models.

Read the notation aloud:
“Coordinates in B ,
mapped into coordinates
in A .”

We can now relate two frames. A robot requires us to connect an entire chain of them.

Composing transformations

Compatible frame labels tell us which maps can be multiplied and which intermediate coordinates disappear.



$${}^A T_C = {}^A T_B {}^B T_C$$

Takeaway: the repeated frame B is the compatible intermediate coordinate system.

For column vectors, the rightmost active motion acts first

The same rotation and translation produce different results when their order is reversed.

$$R_z(90^\circ) T_x(1)$$

First apply T_x ; then R_z rotates both the point and the translation direction.

$$T_x(1) R_z(90^\circ)$$

First apply R_z ; then T_x translates the rotated point along the fixed reference x -axis.

Keep the semantics explicit: active motions move geometry; passive transforms re-express coordinates. Both use matrix multiplication, but the sentence attached to the product must state which interpretation is intended. Frame-index composition still reads as the path $C \rightarrow B \rightarrow A$.

Why we need the D-H convention

D-H stands for **Denavit-Hartenberg**. Its four parameters describe the relative pose between deliberately placed consecutive link frames.

A completely free rigid transform has six degrees of freedom.

$SE(3)$: 3 translations + 3 rotations

That freedom is useful, but it makes a serial robot difficult to tabulate consistently.

D-H chooses consecutive frames around the joint axes and common normal so that only four numbers remain.

$$(a_i, \alpha_i, d_i, \theta_i)$$

Purpose of D-H: a compact, auditable link-by-link table after the joint axes and common normals have fixed the frame structure.

Not unique: parallel, intersecting, or coincident axes can admit several valid origin, sign, and zero-offset choices. Equivalent tables are acceptable when they reproduce the same base-to-tool kinematics.

Original paper: J. Denavit and R. S. Hartenberg (1955), [A Kinematic Notation for Lower-Pair Mechanisms Based on Matrices](#), *Journal of Applied Mechanics*, 22(2), 215–221.

One standard D-H row means four ordered motions

The parameter values and their multiplication order jointly define the relative pose from frame $i - 1$ to frame i .

$${}^{i-1}T_i = R_z(\theta_i) T_z(d_i) T_x(a_i) R_x(\alpha_i)$$

1 · θ_i Rotate about z_{i-1} .	2 · d_i Translate along z_{i-1} .	3 · a_i Translate along the common normal x_i .	4 · α_i Rotate about x_i to align the next z-axis.
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The order is part of the convention. Modified D-H uses a different order and indexing; never combine rows from the two conventions without converting them.

Reading the four D-H parameters

Each parameter measures either separation or alignment between two consecutive joint axes.

$$a_i$$

How far apart are the two joint axes along their common normal?

$$\alpha_i$$

How much must one joint axis twist about the common normal to align with the next?

$$d_i$$

How far do we move along z_{i-1} before reaching the common normal?

$$\theta_i$$

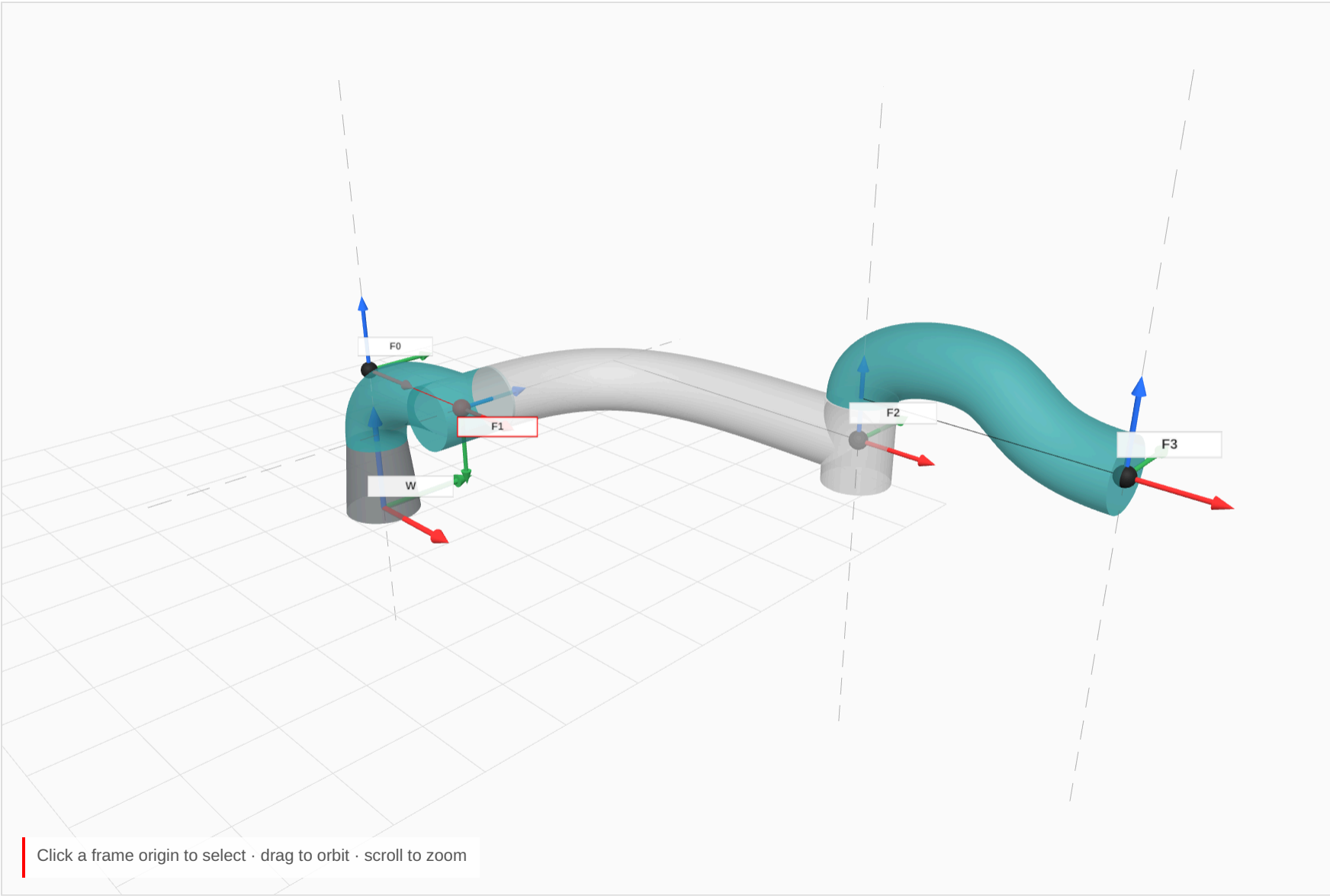
How much do we rotate about z_{i-1} before the common normal is aligned?

Revolute joint: $\theta_i = q_i + \theta_{i,0}$ varies. **Prismatic joint:** $d_i = q_i + d_{i,0}$ varies. Record every zero offset and use consistent angle and length units.

Build and test a D-H frame assignment

Change the four parameters, compare frames, and test whether a selected pair can be represented by one standard D-H row.

Explore: edit θ_i and watch the D-H frames and loaded robot visuals move. Then compare a direct parent–child pair with a nonconsecutive pair.



Click a frame origin to select · drag to orbit · scroll to zoom

▼ Frames

F0

fixed frame

F1

$a\ 1.00 \cdot \alpha\ -90^\circ$

F2

$a\ 2.00 \cdot \alpha\ 90^\circ$

F3

$a\ 1.50 \cdot \alpha\ 0^\circ$

Reset custom_3R

Clear

▼ Robot model

URDF + referenced STL files

Choose Files

No file chosen

Choose one URDF and all of its STL files together. Files stay in this browser.

☒ Show robot meshes

STL opacity 65%

Default: custom_3R URDF with 4 STL visuals and D-H frames F0–F3.

▼ Add / edit with standard D-H

Parent frame

F1

Frame name

F1

Takeaway: D-H parameters describe a constrained relationship between deliberately assigned consecutive frames.

Not every frame pair is one D-H step

A general rigid transform has six degrees of freedom, so an arbitrary pair usually cannot satisfy the four-parameter standard D-H structure exactly.

The playground constructs a standard D-H candidate from ${}^A T_B$ and tests whether it reconstructs the selected transform.

$$\hat{T}_{\text{D-H}} = R_z(\hat{\theta})T_z(\hat{d})T_x(\hat{a})R_x(\hat{\alpha})$$

If reconstruction is not near exact, the selected frame pair does not satisfy the standard D-H placement constraints.

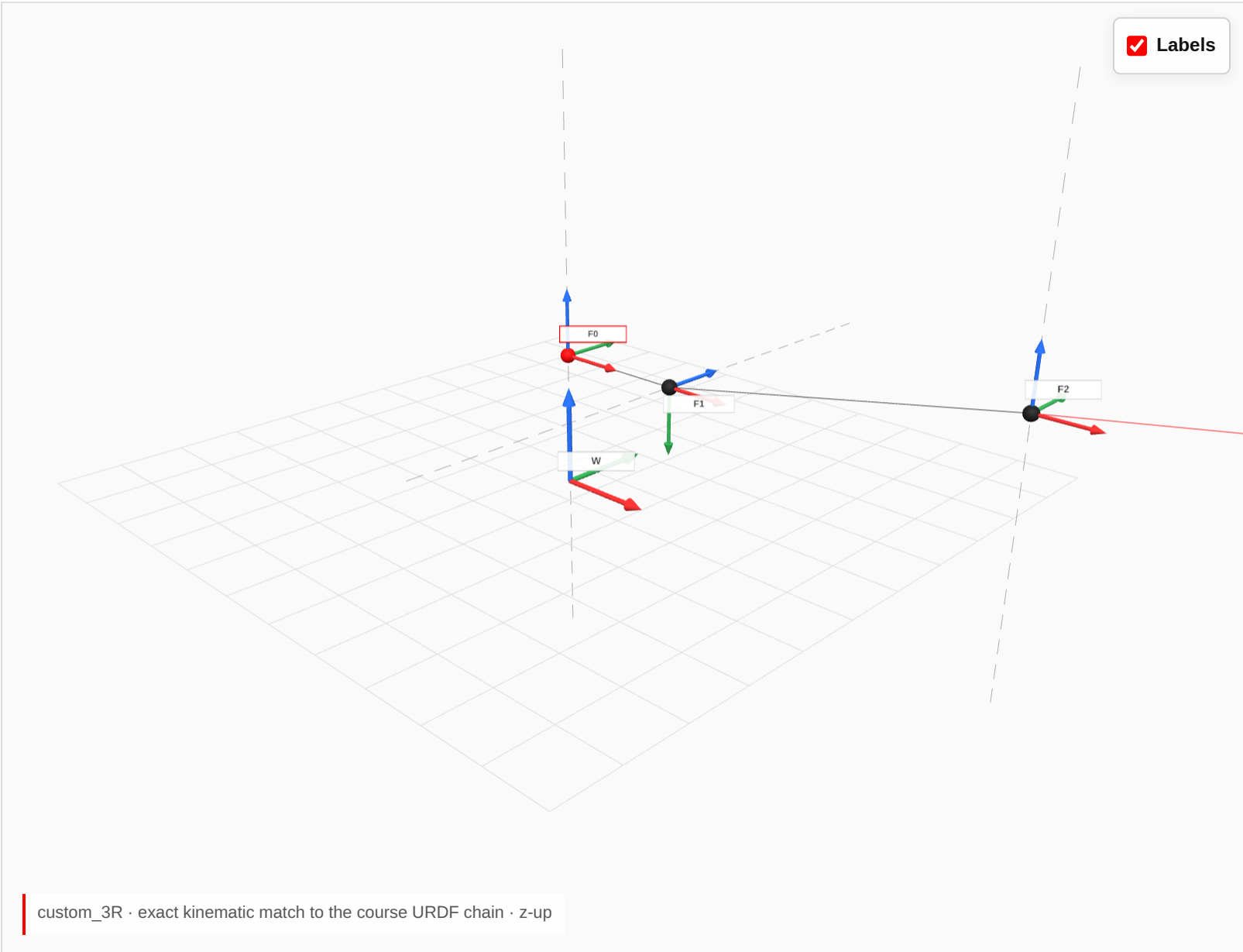
- Test 1:** select a parent and its direct D-H child → reconstruction should be essentially exact.
- Test 2:** select F0 and F2 in the custom_3R preset → two D-H steps generally cannot be compressed into one D-H step.
- Lesson:** D-H parameters belong to a *frame convention*, not merely to two arbitrary coordinate systems.

A scalar matrix residual is only a diagnostic because it mixes dimensionless rotation entries with translation. For model validation, report position error in metres and orientation error in radians separately.

Animate the compact model before comparing representations

This view renders the standard D-H frames and a line model. The dimensions come from the custom_3R URDF geometry, which remains the independent reference model.

Explore: move each joint, identify the active D-H row, and predict how the base-to-tool transform changes. The next step is to compare its pose numerically with the URDF model.



custom_3R · exact kinematic match to the course URDF chain · z-up

Move the three revolute coordinates

q1 0°

q2 0°

q3 0°

Standard D-H table

i	a _i [m]	α _i	d _i [m]	θ _i
1	1.00	−90°	0.00	q ₁
2	2.00	+90°	1.25	q ₂
3	1.50	0°	0.25	q ₃

Fixed base transform: ${}^WT_0 = T_z(1.0\text{ m})$. The D-H frame origin may lie anywhere on the same revolute axis; it need not coincide with the URDF joint-origin point.

Tool pose WT_3

1.000	0.000	0.000	4.500
0.000	1.000	0.000	1.250
0.000	0.000	1.000	1.250
0.000	0.000	0.000	1.000

The compact D-H model

Three standard D-H rows plus explicit base and tool choices reproduce the custom_3R reference pose.

i	a_i [m]	α_i	d_i [m]	θ_i
1	1.00	-90°	0.00	q_1
2	2.00	$+90^\circ$	1.25	q_2
3	1.50	0°	0.25	q_3

$${}^W T_{tool} = T_z(1.0) A_1(q_1) A_2(q_2) A_3(q_3)$$

The D-H origin need not coincide with the URDF joint-origin point; it must lie on the same physical revolute axis and satisfy the chosen convention. A different valid assignment may change the table, base transform, joint zero offsets, or fixed tool transform while preserving the same physical map.

Validate before you trust the model

Equivalent models must produce the same position and orientation—not only at zero, but across representative joint configurations.

For several configurations $q^{(k)}$, compute the tool pose from every representation you claim is equivalent.

$$e_p = \|\mathbf{p}_{\text{D-H}} - \mathbf{p}_{\text{ref}}\|_2 \quad [\text{m}]$$

$$e_R = \cos^{-1} \left(\frac{\text{tr}(R_{\text{ref}}^T R_{\text{D-H}}) - 1}{2} \right) \quad [\text{rad}]$$

Check the zero configuration first.

Then test random joint configurations.

Compare position and orientation with separate tolerances.

Clamp the inverse-cosine argument to $[-1, 1]$ when round-off pushes it just outside the valid range.

Frame-sign and zero-offset mistakes often look plausible until this test.

Use the same physical base and tool frames in both representations.

From physical geometry to a trustworthy model

A reproducible robot model emerges from explicit frame notation, compatible transformation composition, a consistent frame convention, and numerical validation.

- 01** **Coordinates are frame-dependent.** Always state what frame a vector or matrix is expressed in.
- 02** **Homogeneous transforms compose along compatible frame labels.** Order matters.
- 03** **D-H is a frame-assignment convention.** The four parameters are meaningful only after the consecutive frames satisfy that convention.
- 04** **Use the interactive lab as a falsification tool.** A non-zero D-H residual tells you the selected frame pair is not a valid one-step standard D-H relation.
- 05** **Validate numerically.** The next lecture reads the software model in URDF and expresses the same motion through screw axes and PoE.